Mapping Diachronic Change in the Settlement Organization of Sparta: A GIS-Based Approach

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Introduction

This essay presents new insights about and visualizations of the settlement organization of Sparta in the Archaic through Roman periods made possible by what we believe to be the first-ever spatially-enabled database compiling, in quantitative form, all of the published finds from systematic and rescue excavations in Sparta. While compilation of archaeological data is of considerable importance at any site, it is particularly crucial at Sparta. In the remainder of this introduction, we discuss the reasons why that is the case, before briefly laying out the structure of the rest of the essay.

Our knowledge of the physical realities of the ancient city of Sparta has long been sharply circumscribed by the quantity and quality of the relevant evidence. The presence since the 1830s of a modern city directly on top of the ancient remains has hindered archaeological exploration, and systematic excavations have been limited in terms of scope, duration, and publication. Members of the British School at Athens have carried out virtually all of the systematic excavations in Sparta, starting with two campaigns, one lasting from 1906 to 1910, and another lasting from 1924 to 1928, in both cases with limited resources. The 1906-1910 seasons were devoted primarily to tracing the line of the Hellenistic city wall and excavating the Sanctuary of Artemis Orthia. The British team seems to have found the exigencies of dealing with the tens of thousands of small finds from Orthia somewhat overwhelming, and a general sense of disappointment at the lack of monumental architecture and sculpture led to the cessation of excavation after just five years[1]. The dislocations created by World War I (during which Guy Dickins, perhaps the most energetic member of the British team, was killed) delayed final publication of the Artemis Orthia material until 1929, and that publication took the form of a single volume with a little over 400 pages of text – not what one might have hoped given the amount of material in question (Dawkins 1929b). When Arthur Woodward, who dug at Sparta during the 1906-1910 campaign, become director of the British School in 1923, he decided to renew excavations in Sparta, but once again the scope of exploration was limited, with the focus being on the Roman theater and the Sanctuary of Athena Chalkioikos. The results of the five years' of excavations carried out under Woodward are known solely through preliminary reports filed in the *Annual of the British School*.

When Woodward stepped down as director of the British School, the British work in Sparta ceased for an extended period of time, though Richard Nicholls did carry out a brief excavation in 1949 in advance of the construction of a stadium in the city. In 1988, the British School returned to the town of Sparta itself after a nearly 40-year hiatus. During that time, Hector and Richard Catling had excavated the Menelaion, but no work was done by the British in Sparta proper. That changed when Geoffrey Waywell and John Wilkes carried out four seasons of work on the second-century CE structure typically known as the Roman Stoa. When their work on the Roman Stoa was completed, they were invited by then-Ephor Dr. Theodoros Spyropoulos to turn their attention to the theater, where they worked between 1992 and 1998. Further excavations were conducted in the theater in 2007-2008 as a synergasia of the British School and the Greek Archaeological Service with the goal of investigating the late Roman and Byzantine use of the space. Once again the results were published in the form of reports in the *Annual of the British School*.

British archaeologists have thus dug at Sparta only intermittently, concentrated their efforts on a small number of sites – Artemis Orthia, Athena Chalkioikos, the Roman theater, the Roman Stoa – distributed over a limited area, and reported the results of their work rather briefly. For the sake of comparison, it will be helpful to recall that the Deutschen Archäologischen Instituts has been digging in the Kerameikos cemetery at Athens with limited interruptions since 1913 and that the *Kerameikos; Ergebnisse der Ausgrabungen* series presently includes 22 monographs; the American School of Archaeology has been excavating the Agora in Athens more or less continuously since 1931 and the related publication series presently includes 38 monographs.

Furthermore, the literary sources bearing on the settlement organization of Sparta are less informative than one might wish[2]. Spartiates did not produce a great deal of writing, and most of that has not survived to the present day. Non-Spartiate authors occasionally found it expedient to offer some observations on the city of Sparta, but typically only in passing and in brief compass. The *locus classicus* is of course Thucydides' statement that the city of Sparta was visually unimpressive and devoid of remarkable structures and that it was arranged κατὰ κώμας (1.10; Hornblower 1991–2008: vol. 1: 33-5)[3]. The only extant, detailed description of the city, that offered by Pausanias (3.11-18), has proven to be more confusing than enlightening to modern scholars. Nicholas Crosby, writing in 1893, characterized Pausanias' account of the city as 'a most harrowing combination of precision and vagueness', and noted wryly that 'at times he appears almost to be mocking his puzzled readers' (Crosby 1893: 335). There has since Crosby's time been some limited progress in using Pausanias to understand the settlement organization of Sparta in the second century, but that undertaking continues to be hampered by the fact that only a handful of the more than 100 architectural features of the city mentioned by Pausanias have been convincingly identified (see, for example, Gengler 2008). The locations of several major features of the city highlighted by Pausanias, for example the gymnasium complex known as the Dromos, remain unknown (Sanders 2009). Valeria Tosti has in fact argued that Pausanias produced not an accurate description of the city, but rather a largely literary narrative that needs to be read as such (Tosti 2016).

All of this is not, however, a counsel of despair, thanks primarily to work carried out by Greek archaeologists. Members of the Greek Archaeological Service have, starting in the early decades of the 20th century, carried out numerous excavations in Sparta. Some of those excavations have been targeted, short-term digs at specific sites or structures (see, for example, Adamantiou 1931; Adamantiou 1934; Christou 1963), others have taken place as part of longer-term projects of cleaning and conservation (see, for example, Giannakaki 2008). But most of the work in Sparta carried out by the Greek Archaeological Service has taken the form of rescue excavations.

The number of rescue excavations in Sparta was, for a long period of time, quite limited, because only small parts of the city had been designated as archaeologically protected areas that had to be excavated as a preliminary to modern construction projects (see Figure 1). The extension of archaeological protection to the entirety of the area within the Hellenistic city walls, enacted by decrees issued in 1994 and 1995, coincided with a building boom in Sparta, and the result was a sudden and sustained spike in the number of rescue excavations in Sparta, many carried out in portions of modern city that had never been excavated. That shift had profound consequences for our knowledge of the physical realities of ancient Sparta. To give but one example, excavations carried out in 1994 and 2008 uncovered two substantial, previously unknown, cemeteries on the western edge of the city (Christesen 2019, 309-12).

Unfortunately, the nature of the evidence generated by rescue excavations and the fashion in which those excavations are reported have, taken together, limited the usefulness of that evidence for enhancing our understanding of the settlement organization of ancient Sparta. Rescue excavations typically take place within highly restricted spaces that become dig sites not because of careful choices by archaeologists of where it might be productive to excavate, but rather because of decisions of modern-day property owners to initiate a construction project. Rescue excavations thus reveal small, discontinuous patches of the ancient city. The results are reported in the journal Αρχαιολογικόν Δελτίον, which contains a brief (rarely more than two pages), lightly illustrated entry for each rescue excavation carried out in a given year. The members of the Greek Archaeological Service who write those reports are habitually laboring under enormous workloads, and they rarely have the time or opportunity to carefully research and discuss how the finds from the rescue excavation about which they are writing relates to the finds from prior rescue excavations in the same general area.

The evidence from rescue excavations in Sparta is thus both generated and reported in a piecemeal fashion, and its value can only be fully realized when it is assembled into a coherent whole, carefully analyzed, and clearly visualized. All three stages of that process can be most effectively completed through the use of software, typically referred to as a Geographic Information System (GIS).

In the text that follows, we begin with a quick review of the topography of and major landmarks in Sparta and of previous scholarship on the settlement organization of the ancient city. We then briefly discuss what GIS systems are and how they function, before turning to a more in-depth treatment of the foundational work we have done to create both a database of finds from Sparta and the spatial files that make it possible to connect those finds to highly specific locations. Finally, making use of the database and files, we offer some insights into the settlement organization of Sparta in the Archaic through Hellenistic periods and how it evolved over the course of time. We pay particular attention to burials, cult sites, and craft production sites (the last of these as an indicator of the extent of the densely settled part of the city).

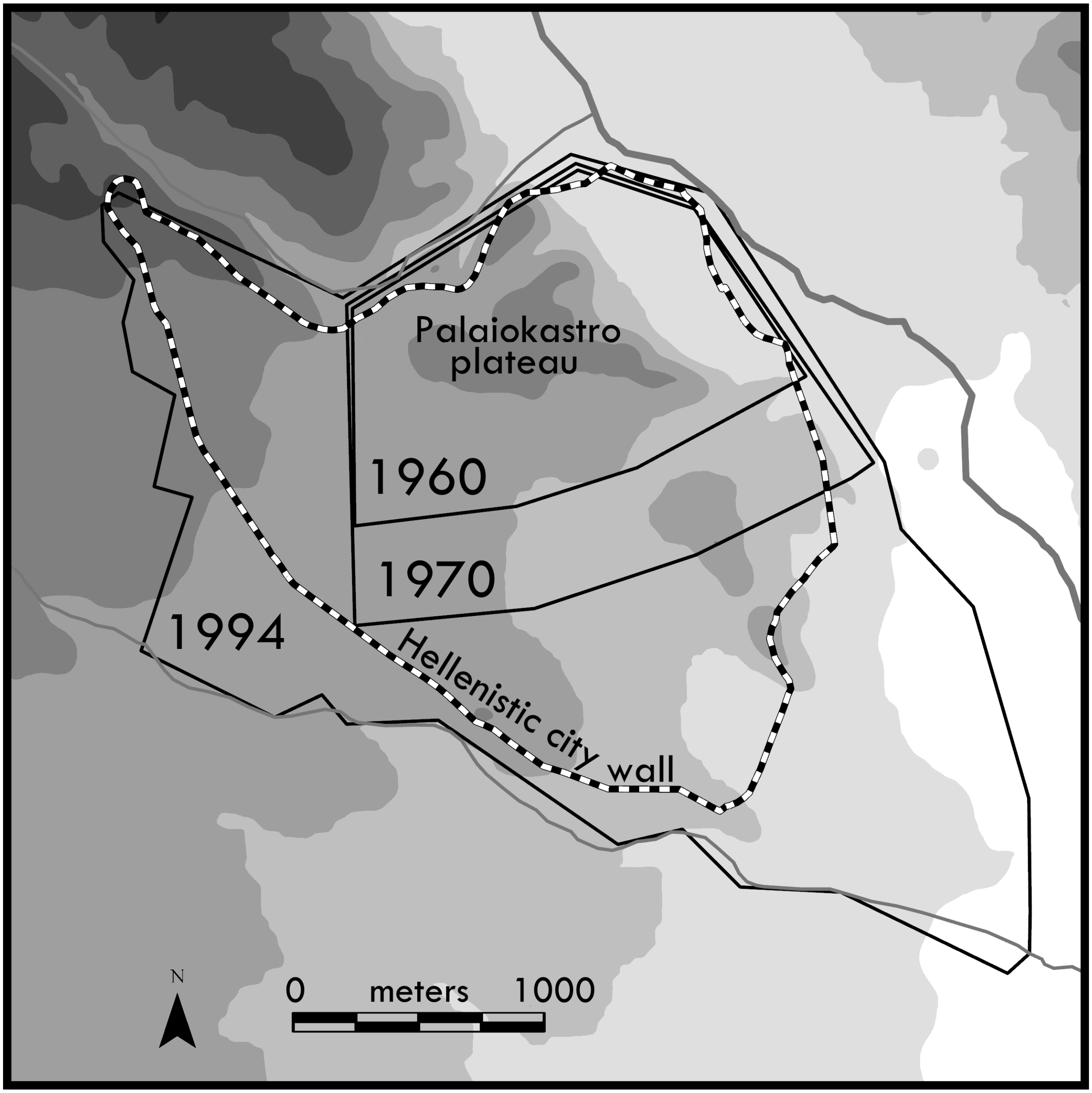

Figure 1: Changes in the extent of archaeological protection in Sparta

Topography and major landmarks

The site of Sparta is bounded to the north by the Mousga river and ravine, to the south and southwest by the Magoulitsa river and ravine, and the Eurotas to the east (see Figure 2)[4]. The two most important topographic features inside the city are the acropolis and adjoining Palaiokastro plateau. The city stretched south from the Palaiokastro plateau, in the space defined by the Mousga, Magoulitsa, and Eurotas. A series of isolated hills, though not much higher than the surrounding ground level, had sufficiently steep sides as to be significant factors in the topography of the city and in channeling movement in the city. These hills include Gerokomeiou and Xenia, which are located just to the south of the Palaiokastro plateau, and Evangelistria, alongside the Magoulitsa.

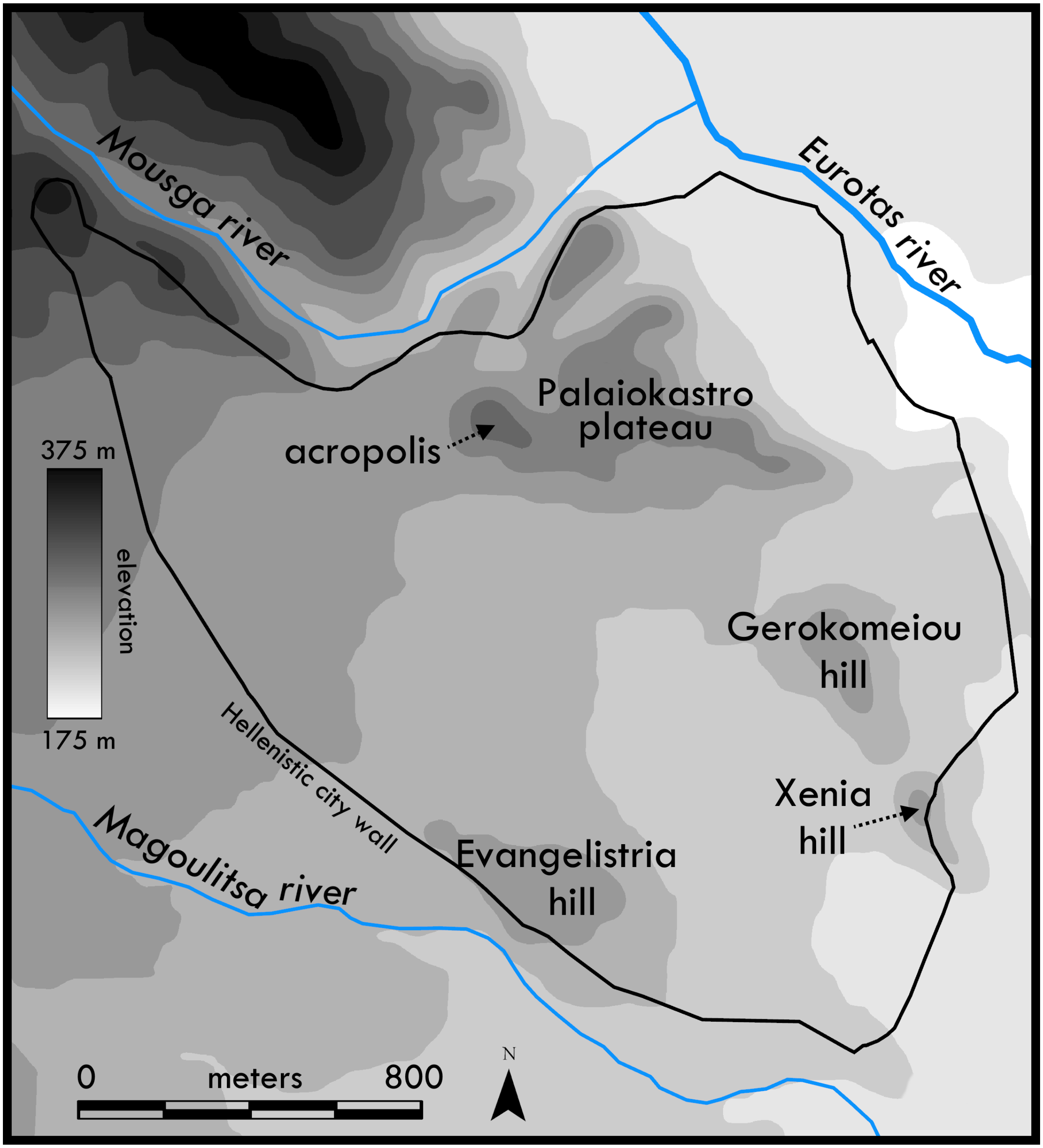

Figure 2: Topography of Sparta

A fortification wall (#1 on Figure 3) was constructed in the Hellenistic period, and another, smaller fortification (typically called the Late Roman fortification wall; #2 on Figure 3) was constructed around the acropolis and the Palaiokastro plateau sometime in the fourth century CE (Wace 1907; Frey 2016, 85-127). This later wall enclosed numerous earlier buildings including the Sanctuary of Athena Chalkioikos (#3), a theater built in the Augustan period (#4), an Archaic

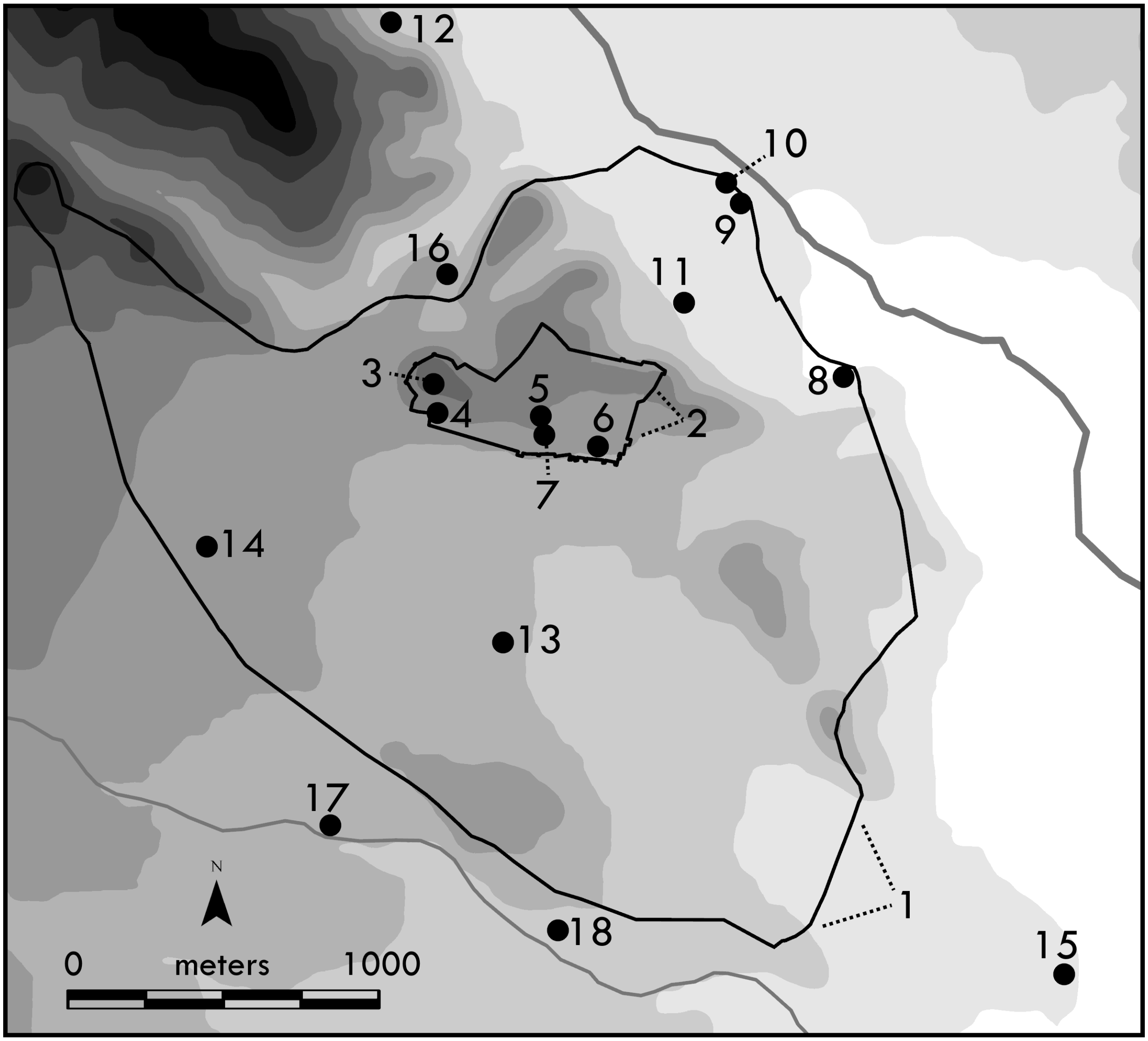

Figure 3: Landmarks in Sparta

stoa (sometimes called Christou's stoa after the excavator; #5), a second-century CE stoa (typically called the Roman stoa, #6), and a circular terrace wall built in the Archaic period (typically called the Round Building, #7)[5].

Several cult sites were located on the west bank of the Eurotas, including the Sanctuary of Artemis Orthia (#8), a heroon (frequently referred to as the Heroon of Astrabakos or the Astrabakos monument, #9), a stone platform (typically referred as the Eurotas or Lycurgus Altar, #10), and the Stavropoulos sanctuary (named after the former owner of the plot in which it is located, #11). To the north of these sites, outside the city walls and on the route to/from Arcadia, lies another sanctuary (typically referred to as the Achilleion, #12)[6].

To the south of the acropolis is a small structure built from massive blocks of conglomerate (typically called the Leonidaion because of a probably mistaken belief that it is the tomb of King Leonidas, #13). A few hundred meters to the west is a Roman bathing complex (typically called the Arapissa baths, because they were associated with a so-named heroine in local legend, #14). To the southeast of the city, near the convergence of the Magoulitsa and Eurotas rivers, are the remains of a large Roman burial monument (typically called the Altar of Psychiko, #15)[7].

Three extra-communal cemeteries have been identified at the fringes of Sparta: one in the Mousga ravine (#16) and two (what are typically called the Southwestern Cemetery and Olive Oil Cemetery (#17, 18)) along the Magoulitsa (Christesen 2019).

Previous scholarship

Serious discussion of the topography of Sparta began just prior to the construction of modern Sparti, with observations on the visible remains made by W.M. Leake and the French Morea Expedition in the early decades of the 19th century (Leake 1830: vol. 1: 124-87; vol. 2: 532-4; vol. 3: 57-83; Blouet 1833, 61-83). The British excavations that began in 1906 resulted in major advances of our understanding of the settlement organization of ancient Sparta. The British team devoted a considerable amount of energy to establishing some basic facts about the layout of the ancient city, for example by carefully tracing the course of the Hellenistic and Late Roman fortification walls and definitively establishing the locations of the sanctuaries dedicated to Artemis Orthia and Athena Chalkioikos. The plan produced by Wassily Sejk as part of the British work in Sparta from 1906 to 1910 has long remained the definitive map of the city (Wace 1907: Pl. 1). However, the fact that the British campaigns of 1906-1910 and 1924-1928 left much of the city unexplored meant there were still big gaps in our understanding of Sparta's settlement organization.

Much of the rest of the 20th century was an extended fallow period with respect to studies of the settlement organization of Sparta. In the absence of new archaeological data, the relevant scholarship relied on the same collection of evidence and consisted largely of attempts at connecting the available archaeological data with Pausanias' description of the city (see, for example, Stibbe 1989).

As one would expect, the series of rescue excavations that began in 1994 produced an ever-accumulating collection of new evidence for the settlement organization of Sparta. Excavations took place in parts of the city that had never previously been archaeologically excavated, and it was possible for the first time to fill in many blank spots on plans of ancient Sparta.

The most immediate and important resulting piece of scholarship on the settlement organization of Sparta was Eleni Kourinou's monograph of the topography of the city that was published in 2000 (Kourinou 2000). Kourinou drew on new archaeological and epigraphic evidence, along with the long-known literary sources, in discussions of the city's fortification walls, bridges, *ōbai*, acropolis, agora, road network, sanctuaries, cemeteries, and water supply. As Graham Shipley notes in his review of Kourinou's book, 'it represents the most important contribution to the topography of Sparta since the British campaigns of the early 20th century' (Shipley 2003, 132).

Nonetheless, Kourinou's work is now somewhat dated, in two different ways. First, literally hundreds of rescue excavations have been carried out in Sparta since 2000, and thus we now have access to a great deal of new evidence. Second, Kourinou's book was written without any use of geospatial software, hence the archaeological evidence is treated in a qualitative rather than quantitative fashion, and Kourinou produced just one (less than highly polished) map to accompany the text. Moreover, Kourinou's monograph is for all intents and purposes synchronic – there is little attempt to trace changes in the urban fabric of Sparta over time.

Numerous, typically brief, articles on the settlement organization of Sparta have appeared in the past 20 years, many of them written by Greek archaeologists working in the city. However, the scholars responsible for this work have relied primarily on data from sites they themselves excavated, thus limiting the scope of their inquiry (see, for example, Vasilogambrou, Tsouli and Maltezou 2018). Other scholars have been more eclectic in their collection of evidence but focused on a single subset of sites inside Sparta, namely the sanctuaries and votive deposits (see, for example, Pavlides 2011). There is also a continuing trickle of publications that rely primarily on Pausanias to reconstruct the layout of Sparta, with a particular focus on the location of the agora (the part of the ancient city that is described in the most detail by Pausanias; see, for example, Greco 2016).

There has, as a result, been a major gap in the scholarship on the settlement organization of ancient Sparta in that the growing body of archaeological evidence has not been utilized nearly as fully as it might. That gap was partially filled by an article published by one of the present authors on the typology and topography of burials in Sparta (Christesen 2019). However, the analysis in that article is limited to just one category of evidence, and does not leverage the capacities of a GIS system. This article represents a step forward because it encompasses all categories of evidence and makes use of the analytical and visualization capacities of geospatial software.

What is GIS?

A Geographic Information System (GIS) system is a software package that stores and displays data that has a spatial component. GIS software is produced by a number of different companies, but the two most common versions are ArcGIS Pro (a commercial product offered by a company called ESRI) and QGIS (free shareware created by the Open Geospatial Foundation). The work described in this article was done in ArcGIS Pro[8].

A GIS project is built around a database of objects that have two features: (1) a specific location and (2) one or more attributes. The GIS software stores information about both features – location and attributes – of each object and allows users to analyze and display that data in a variety of different ways.

GIS databases are arranged in a tabular format (i.e. in rows and columns like a typical spreadsheet) and are structured such that each row contains information about a specific location, and each column contains data about a specific attribute. A very simple GIS database could, for example, contain details about the houses in a specific neighborhood (see Table 1).

Table 1: Highly simplified GIS database

address	Latitude	Longitude	Year Built	Total # of Rooms
3 Balch Street	43.705324	-72.280390	1922	9

5 Balch Street	43.705588	-72.280447	1938	10
7 Balch Street	43.705913	-72.280487	1932	12
9 Balch Street	43.706276	-72.280637	1930	9
11 Balch Street	43.706551	-72.280696	1927	8

A complex GIS database can contain separate entries for millions of locations and hundreds of pieces of information about each of those locations.

The visualization of data in a GIS system is greatly aided by the fact that a number of different types of data can be placed on top of each other in separate layers, each of which can be turned on and off. In addition, data can be displayed using toggleable symbology to differentiate and display features such as vase type, vase date, and vase painter independently.

<u>Foundational Work: Attribute Data</u>

As noted above, we have constructed what we believe to be the first spatially-enabled database that compiles, in quantitative form, all of the published finds from systematic and rescue excavations in Sparta. The initial stages of data entry were done in Excel, and the resulting spreadsheet was then transitioned into a Postgres database.

The majority of the information in the database is derived from entries in the Αρχαιολογικόν Δελτίον that report the results of rescue excavations in Sparta. We compiled data from the Δελτίον up through and including Volume 69 (2014). The extension of archaeological protection in 1994-1995 resulted in a major upward shift of the number of rescue excavations, as is apparent from Table 2. In addition, we made use of information from the excavation reports by the British excavators who have worked in Sparta and from articles written by members of the Greek Archaeological Service discussing finds from excavations in Sparta that are not reported in the Δελτίον (either because they are slated for inclusion in a volume of the Δελτίον that has not as yet appeared or because the excavation in question was not otherwise reported).

Table 2: Number of rescue excavations in Sparta reported in the Αρχαιολογικόν Δελτίον

Years	Number of Excavations
1960s	19
1970s	26
1980s	42
1990s	165
2000s	90
2010s	135

Entries in the Δελτίον, especially the most recent (post-1990) reports, are fairly standardized in their presentation. They consist of four main parts:
(1) A title, which typically supplies the name of the owner of the plot of land in which the excavation was conducted and frequently other location information, such as a building block number or, more rarely, a street address.
(2) A description of the built features uncovered during excavation, including wall dimensions, stratigraphy, etc.
(3) Moveable finds, sometimes with precise numerical values, sometimes with imprecise quantitative adjectives (‘a few’, ‘some’, ‘many’, etc.).
(4) Plans and photographs.

The information in any given Δελτίον entry needs to be inputted into the database through a process called encoding. A key issue is that Geographic Information Systems, like most computer programs, are strongly optimized for consistently formatted quantitative data and are not well adapted to storing or analyzing inconsistently formatted qualitative data. It is, as a result, necessary to transform qualitative descriptions such as ‘much pottery of the Archaic period’ into quantitative information (see below for further discussion).

The database is divided into four linked and related sections: bibliographic information, location information, finds (including functional designations), and burials. Each row in the bibliographic, location, and finds sections contains information from a specific excavation report and corresponding site. If multiple excavations have been carried out at the same place, there are multiple rows for that site. The burials section contains one row for each burial (and hence, for example, a cluster of six burials at the same site would occupy six rows in the database). Each excavation report is assigned a unique 19-digit number that encodes information about the location of the site in question and the year in which the excavation took place. That 19-digit number functions as a key field by means of which the four sections of the database are linked together.

The bibliographic information requires no special explanation, and the location information is discussed in detail below. Find data is reported under 42 separate headings (see Table 3); each heading occupies multiple columns in the database because both quantitative and chronological information is included.

Table 3: Find categories in database

architecture	coins	pipes
akroteria and antefixes	domestic utensils	pottery
amphoras	glass objects	relief pithoi
animal bones	hearths	roads
architectural reliefs	inscriptions	roof tiles
architectural sculpture	iron objects	seals
bells	jewelry	statuary
bone carvings	lead figurines	stone objects
bone figurines	loom weights	stone reliefs
bone objects	miniature pottery	terracotta figurines
bronze figurines	mosaics	terracotta masks
bronze objects	musical instruments	terracotta plaques
bronze vessels	perirrhanteria	weapons
burials	pins	wells

The quantities of built features – walls, roads, pipes, mosaics, etc. – are, for the most part, reported accurately and consistently. For the moveable finds, which range from akroteria to weapons, different reports will mention ‘some’, ‘lots’, ‘many’, ‘a few’, etc. What these subjective words mean in objective numerical values is impossible to judge. Therefore, to reduce inconsistency, we created categories of qualitative descriptions with corresponding numbers (see Table 4).

Table 4 : Quantitative assessments of qualitative descriptions

number inputted into database	pottery	ubiquitous items (e.g. figurines, miniature pottery)	less frequent items (e.g. terracotta plaques)	rare items (e.g. statuary.)
5.1	n/a	‘a few’	‘some’	‘were found’
15.1	‘a little’	‘a good amount’	‘fragments’	‘many items’
50.1	‘was found’	‘many’	‘a large amount’	a hoard
250.1	‘much’	a hoard	a hoard	N/A

In order to separate precisely reported figures (e.g., ‘50 terracotta figurines’) in a Δελτίον entry from figures that generated based on a qualitative description (e.g. ‘many terracotta figurines’), we added .1 to the end of each of the latter. This makes it possible for anyone using the database to discern whether a particular figure is precise or an approximation.

In recording the dates of built features and moveable finds, we have, except in a few instances, deferred to the expertise of the excavators and carefully recorded any chronological information provided in any given Δελτίον entry. In most cases dates in Δελτίον entries are given by historical period (e.g. Archaic, Classical, Hellenistic, etc.), but in some cases dates are given by century or even decade. In order to keep the chronological information easily legible to users of the database and readable by the GIS system, we created a series of field codes, which is to say two-letter abbreviations for specific periods, as indicated in Table 5 below.

Table 5: Historical period date codes used in the database

field code	period		field code	period
EH	Early Helladic		AR	Archaic
MH	Middle Helladic		CL	Classical
LH	Late Helladic		HL	Hellenistic
SM	Sub-Mycenaean		RO	Roman
PG	Protogeometric		BZ	Byzantine
GM	Geometric		UD	undated

The Δελτίον in some cases provided no dates for a specific built feature or moveable finds, but it was possible to arrive at a date based on the history of a particular category of built feature or object. Terracotta disk akroteria, for example, were produced in Laconia only during the Archaic period (Winter 1990), so if a Δελτίον entry records the discovery of pieces of a terracotta disk akroterion, but gives no date, it is still possible to position that object in time.

The find section of the database also includes information about the function of the site in question. We drew in large part on the functional categories used in the Dipylon Project's Mapping Ancient Athens[9], which are: (1) Public Space, (2) Religion/Cult, (3) Transport, (4) Fortification, (5) Water Supply/Drainage, (6) Domestic Space, (7) Production, (8) Commerce, (9) Funerary Space, (10) Health/Welfare, (11) Education, (12) Unknown. We changed the function of 'health/welfare' to 'bathing' as this was the only type of health or welfare establishment found so far in Sparta, and we have not as yet identified any space in Sparta that can be assigned to the Education heading. We assigned functional categories to specific sites on the basis of the published finds.

Foundational Work: Spatial Reference Systems

In order to understand how locations are recorded in the database, it is first necessary to understand the various spatial reference systems for Sparta that existed prior to this project and the one that we created. There were four separate pre-existing references systems: numbered building blocks, plots of lands within each building block, street names and numbers, and the reference grid created by the British excavators in the early 20th century.

When the modern city of Sparta was built in the 1830s, an urban plan was drawn up that created 200 numbered building blocks, some of which were perfectly square and of uniform size (150 m on a side), while others were irregularly shaped and of varying size (see Figure 4). Some building blocks were divided into sub-sections, each of which was indicated with a letter (e.g. Building Block 123, 123A).

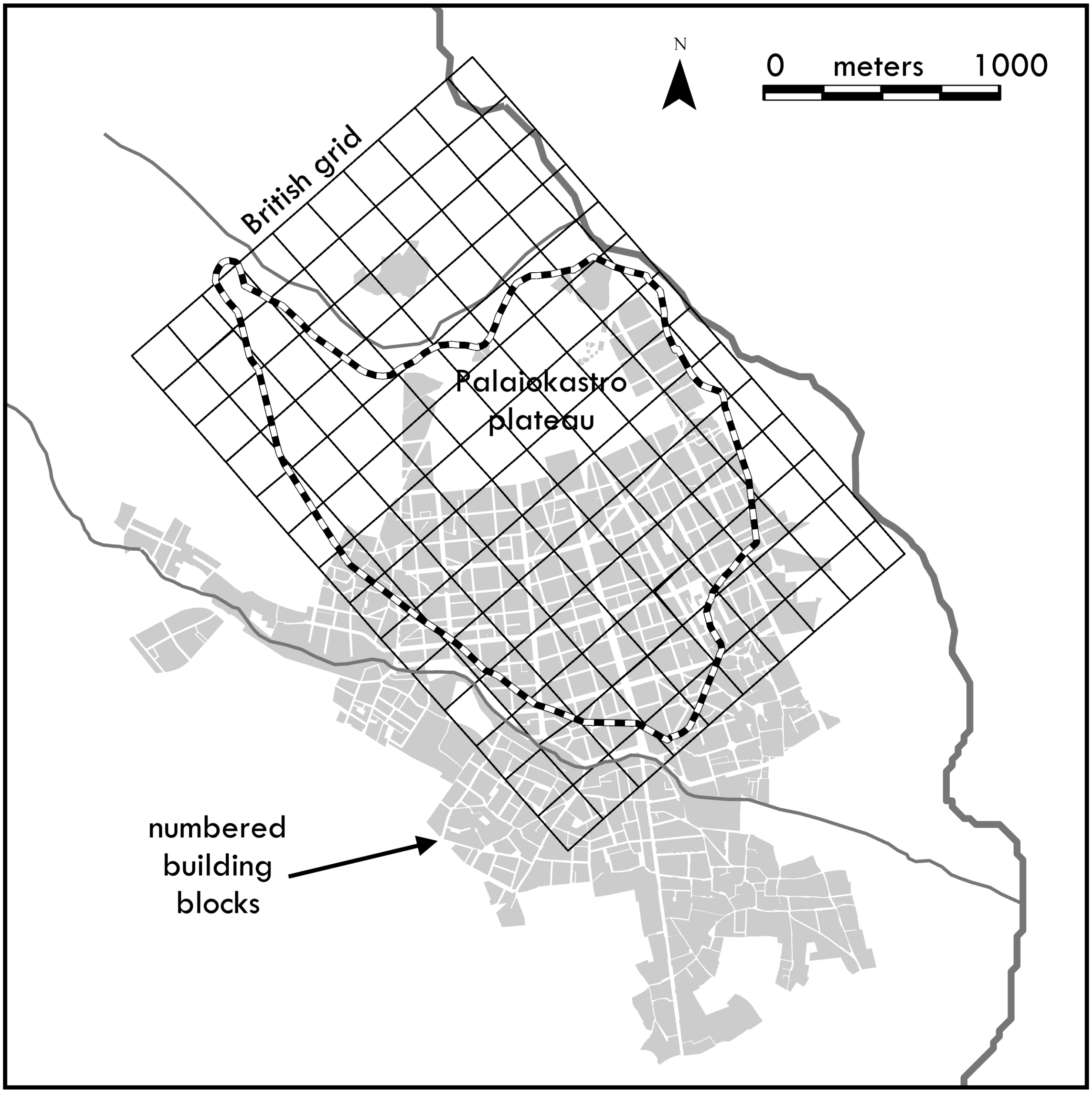

Figure 4: Building Blocks and the British Grid in Sparta

As the city expanded outside of the original 200 blocks, planners created new building blocks and labeled them differently. The next 240 blocks are designated by a Γ and are non-consecutively numbered, meaning that there is Γ17 but not Γ50 and the assigned numbers reach as high as Γ470. There is an additional tranche of building blocks numbered Π3-Π10 that lie on the Gerokomeiou and Xenia hills.

Each building block in Sparta is subdivided into a varying number of plots of land, with each plot varying widely (one might almost say wildly) in size and shape (see Figure 5). As one would expect, each street in the city is assigned a name, and buildings on each street have unique numbers.

Figure 5: Plots of Land within Building Blocks in Central Sparta

When the British excavators arrived in Sparta in 1906, they created a reference grid based on squares that were 200-m on a side and with a NW-SE orientation that was roughly aligned with the course of the Eurotas river. There are 10 squares in the E-W direction (lettered F-P, with I skipped), and 14 squares in the N-S direction (numbered 7-14) (See Figure 4).

Most of the Δελτίον entries for Sparta use one or more of these four systems to report the location of the excavation in question, with the most typical approach being to give building block number and the owner of the plot of land within the building block. For excavations conducted outside the area covered by the building blocks, a British grid reference (e.g. P14) is usually cited. Street names are reported intermittently and street numbers only occasionally.

An important exception is entries pertaining to excavations carried out in streets, which are by definition situated between building blocks. The locations of those excavations are typically given by street name, sometimes with the additional specification of the range of street numbers (e.g. 28-42 Odos Thermopylai) and/or the numbers of the building blocks on either side.
It was, for the most part, unproblematic to enter into the database the locational data for the building blocks, street addresses, and British grids, in large part because some Δελτίον volumes include plans of the entire city showing some or all of the building blocks and their numbers.

The plots of land within each building block presented special challenges. One of the authors of this article had made repeated inquiries over the years in an attempt to locate a map showing the location and owners of plots of land in Sparta, to no avail. That changed in December, 2022 as the result of an ongoing program by the Greek government to carry out a full cadastral survey of the entire country[10]. That survey is a work in progress, but in December we managed to acquire from the Greek government a GIS file showing every distinct plot of land in Sparta. That was an important development because it created the potential to map the location of hundreds of rescue excavations with a much higher degree of precision than was possible if one relied solely on building block numbers or British grid references.

Unfortunately, the GIS file from the Greek government did not include the name of the plot owners. This was a major problem because the Δελτίον entries reference plots by the name of owners, which meant that it was not immediately possible to use the information about the plot in a Δελτίον entry to locate that plot on the map. We addressed that problem by employing a variety of tools to assign plots to owners, and we were able to link the vast majority of the plots named in Δελτίον entries to specific plots of land in Sparta, with the caveat that we could be more confident in some cases than others.

In the course of carrying out this project, it became apparent that none of the existing reference systems was entirely satisfactory. Neither the numbered building blocks nor the British grid cover the entire city, and so they cannot be used to locate many excavations. Street address information for many excavations is either imprecise or unknown. The plots vary widely in size and shape, which creates difficulty in producing legible visualizations.

What was needed was a reference grid of uniformly-sized, small tiles that covered the entire city. To that end, we, with the invaluable help of Dartmouth College's GIS Specialist Stephen Gaughan, created a new grid system. In order to facilitate mapping, this grid is anchored to the geodectic reference system created by the Greek government called the Greek grid, which includes both a datum (a model of the shape of the earth) called GGRS87, and a projection (a mathematical system for representing the three-dimensional space of the earth on a two-dimensional map), transverse Mercator. The Greek grid covers the entire country of Greece, and is built around a specific point (38.078400°N, 23.932939°E) near the Dionysos Satellite Observatory northeast of Athens.

The grid system we created has four layers of overlapping tiles. The first layer consists of a hundred 100 km x 100 km tiles. The next layer consists of 10,000 10 km x 10 km tiles. The third layer consists of 1 million 1 km x 1 km tiles. The final layer consists of 100 million 100 m x 100 m tiles. 100 x 100 m tiles seemed to be the perfect size for plotting the locations of finds from Sparta, as they are a smaller than a standard building block yet big enough to report data without compromising it[11].

Each of these tiles is assigned a unique identifying number, with the numeration starting from the southwest corner and proceeding eastward to the end of the row before moving to the western edge of the next row to the north. For our purposes in Sparta, we are dealing with 100 x 100 km tile number 048, two 10 x 10 km tiles numbered 048.006 and 048.007, 36 1 x 1 km tiles, and 3,600 100 x 100 m tiles (of which only a fraction have any reported finds).

The database as constructed supplies, for each excavation reported in the Δελτίον, a location based on multiple different reference systems (each reported in a separate column):
(1) building block or British grid reference (with street digs assigned to the nearest building block);
(2) street address (where specified in a Δελτίον entry or where it can be established using sources such as satellite imagery);
(3) name of owner of building plot;
(4) KAEK (the unique 12-digit number assigned by the Greek government to each plot of land covered in the cadastral survey);
(5) tile number (based on the system of tiles described above).

In order to accommodate the 100 plus digs that have been conducted on the streets of Sparta, which are on public land that is not specifically sub-divided in the cadastral survey, we created, in the GIS system, new polygons delineating the space covered in any given street dig.

An Example of Spatial Analysis: Clustering of Burials and Moran's Index
GIS systems include tools with capacities to carry out varied and sophisticated data analyses. It is impossible in the present context to discuss all of the ways in those tools could be applied to the database described above, so in this section of the article we explore, *exempli gratia*, how GIS can be used to address two questions about the distribution of burials in Sparta.

The burials section of the database currently contains entries for 408 tombs ranging in date from the Early Helladic to the Roman period. This does not include tombs in Southwestern and Olive Oil cemeteries (see below) but does include tombs in the cemetery in the Mousga ravine[12]. The chronological breakdown is shown in Table 6.

Table 6: Numbers of cataloged burials, by period

EH	MH	LH	PG	GM	AR	CL	HL	RO	UD
2	8	1	15	19	29	17	77	234	6

This data can be analyzed to address two questions:
(1) The current scholarly consensus is that five villages or *ōbai* formed the core of the Lacedaemonian polity, with four of those villages (Kynosoura, Limnai, Mesoa, Pitana) encircling the Palaiokastro plateau (Kourinou 2000, 35-66, 89-95). Kourinou and others have argued that there were from an early date four distinct cemeteries in the inhabited area Sparta, one for each of the *ōbai*. Maria Tsouli has argued that the Olive Oil Cemetery functioned as the necropolis for the *ōba* of Mesoa (Kourinou 2000, 215-19; Tsouli 2013a, 153). The existence of distinct cemeteries tied to specific *ōbai* was called into question in an previous publication by one of the present authors (Christesen 2019, 336-8), but the relevant argumentation was based on a purely visual inspection of some relatively simple maps. When the burials are subjected to a GIS-based analysis, are there clear clusters that would confirm or falsify the existence of four distinct cemeteries in the city itself?

(2) The Spartans were unusual in burying their dead both within the city itself (=intra-communal burial) and just outside the city, in big cemeteries (=extra-communal burials). The area occupied by the city of Sparta became progressively more densely inhabited over the course of time. Did that affect the spatial patterning of burials? Were burials in Sparta more highly clustered in later periods?

We generated answers to those questions by calculating global and local Moran's Indices (frequently referred to as Moran's I) for the published burials in Sparta using as a spatial reference system the 100-meter square tiles discussed above. Spatially-distributed variables (e.g. the locations of instances of a disease) can be spatially clustered, evenly dispersed, or random (see Figure 6 below). In practice, few real-world variables are spatially distributed such that they correspond precisely to the left or middle cube. As a result, the spatial distribution of variables is typically calculated using what is called Moran's Index, which provides a measure of what geographers called 'spatial autocorrelation'. A perfectly clustered variable (left cube) would get a Moran's Index score of 1, a perfectly dispersed variable (middle cube) would get a score of -1, and a completely random distribution would get a score of 0.

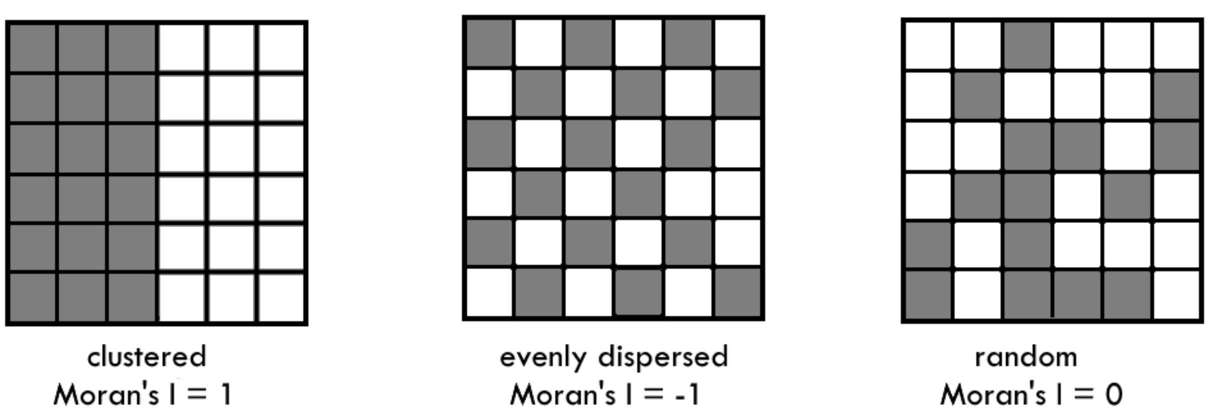

Figure 6: Types of spatial distribution

Moran's Index can be calculated in two different fashions. A global Moran's Index calculation generates solely a numerical score for the dataset in question. A local Moran's Index calculation

identifies the locations of statistically significant clusters in the dataset. We calculated global and local Moran's indices for two different datasets:
(a) dated, intra-communal burials, which we defined as burials within the space defined by the Hellenistic wall circuit (402 burials in all);
(b) intra-communal burials plus burials made within 400 meters or less outside the Hellenistic wall (1,472 burials in all). This calculation included burials in the Southwestern and Olive Oil cemeteries. For analytical purposes we set the number of burials in Southwestern cemetery to 1,000, dated them all to the Roman period, and distributed them over four 100-meter tiles, and we set the number of burials in the Olive Oil Cemetery to 70 (dated as 25 Archaic, 25 Classical, 20 Hellenistic), and placed them all in a single 100-meter tile[13]. In both cases we combined Protogeometric and Geometric burials into a single category because it has proven difficult in some cases to differentiate the tombs of those two periods.

The results of the global Moran's Index calculations are shown in Table 7. The scores for intracommunal burials do not support the hypothesis that there were four distinct cemeteries within the inhabited area of Sparta. There is no trace of clustering in the Archaic, Classical, and Hellenistic periods, and only a limited degree of clustering in the Protogeometric-Geometric and Roman periods. The spatial patterns underlying this data are made clear by the results of the local Moran's Index calculation, which can be found in Figure 7. Burials are represented by proportionally-sized dots, and statistically-significant clusters are shaded in black. In the Protogeometric-Geometric period there was a notable cluster (spanning four tiles) in the northeastern part of the city, and in the Roman period there were clusters in the center and southeastern part of the city[14]. The scores for intra-communal and extra-communal burials taken together tell largely the same story, and also suggest that burials in Sparta did ultimately become significantly more highly clustered, but not before the Roman period.

Table 7: Global Moran's Index scores for burials in Sparta

period	Global Moran's Index	probability that this pattern could be the result of random chance		Global Moran's Index	probability that this pattern could be the result of random chance
	intra-communal burials			intra- and extra-communal burials	
Protogeometric-Geometric	0.15	< 1%		0.15	< 1%
Archaic	-0.01	pattern appears to be random		0.00	pattern appears to be random
Classical	0.03	pattern appears to be random		0.00	pattern appears to be random
Hellenistic	0.04	pattern appears to be random		0.02	pattern appears to be random
Roman	0.15	< 1%		0.49	< 1%

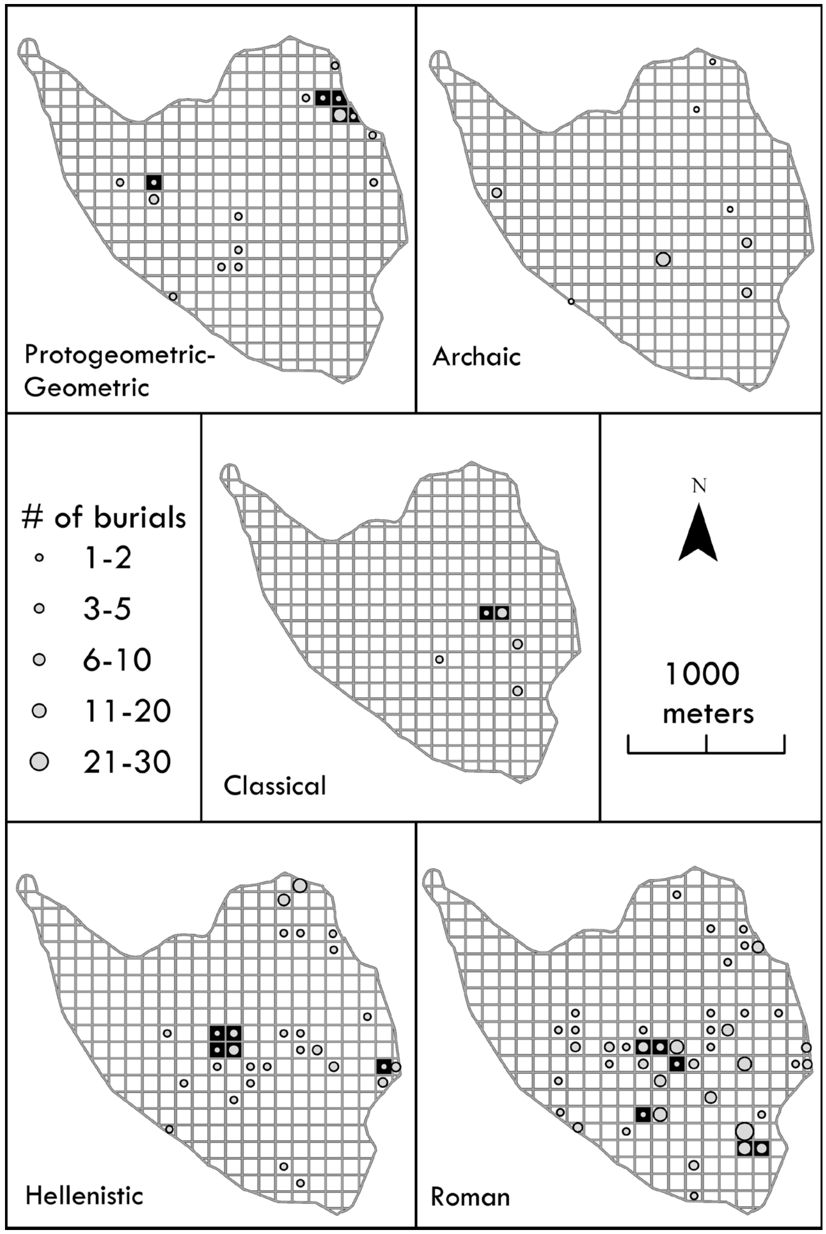

Figure 7: Locations and Clustering of Burials in Sparta, by Period

Cult Sites in Sparta

In the absence of large quantities of published data from cemeteries and public spaces such as the agora, cult sites represent by far the single most important find context in Sparta. The finds from those cult sites, therefore, represent an obvious focal point for analysis of the information in the database and its ramifications for our understanding of the settlement organization of Sparta. We will consider diachronic patterns in the number of cult sites before examining their spatial distribution and the relative weighting of heroic vs. Olympian sanctuaries. We would like to emphasize that our goal here is to simply highlight some important facets of the data, fuller analyses of which will be presented in forthcoming publications.

In the picture emerging from our database, perhaps the single most striking feature of Sparta is the sheer number of sites that have produced evidence for cultic activity (see Figure 8). In Athens, for example, beyond the centers of cultic activity at the Agora and Acropolis only 18 unique religious sites have been identified that were active at any point in the Archaic through Roman periods (as per the Mapping Ancient Athens data). Sparta, by contrast, has 63 sites with some sort of religious function during those periods. To help put that number into perspective, 155 of the 100-m tiles have attested activity of some kind at some point in time between the Archaic and Roman periods; among those 155 tiles, 49 of them have evidence for cult activity. The disparity between Athens and Sparta is even greater if we concentrate on cult sites active during the Archaic period (see Table 8).

Some caution is in order with respect to the figures from Sparta, because much of the evidence for religious activity derives not from architecture but from votive deposits. It is likely that in some instances multiple votive deposits came from a single cult site. However, even if the number of cult sites were reduced by a quarter or even half, Sparta still retains an unusually high density of religious sites.

The decline in the number of cult sites over time is also noteworthy. This is, prima facie, the opposite of what one might expect in Sparta, where the Hellenistic and Roman strata are better preserved and more regularly excavated than the older, deeper strata. The apparent decline can be explained in part by shifts in dedicatory practices. The quantity of objects dedicated in sanctuaries across the Greek world declined markedly after the end of the Archaic period, for reasons that remain less than entirely unclear (see, for instance, Snodgrass 1989–1990; Loy and Slawisch 2021). In addition, dedications of the most important categories of objects that signal the presence of a religious sanctuary in Sparta, lead figurines and terracotta plaques, tapered off dramatically then ceased entirely. The date of the latest lead votives continues to be debated, but there can be little doubt that their numbers plummeted after the early fifth century (Wace 1929; Boss 2000; Lloyd 2024); the latest terracotta plaques from Sparta seem to date to the late fourth century (Salapata 2014). Finally, the dating of the black-glazed pottery that became the standard ware produced in Laconia after the late sixth century can be difficult to date, which makes later cult sites more difficult to trace (see, for example, Bonias 1998, 55).

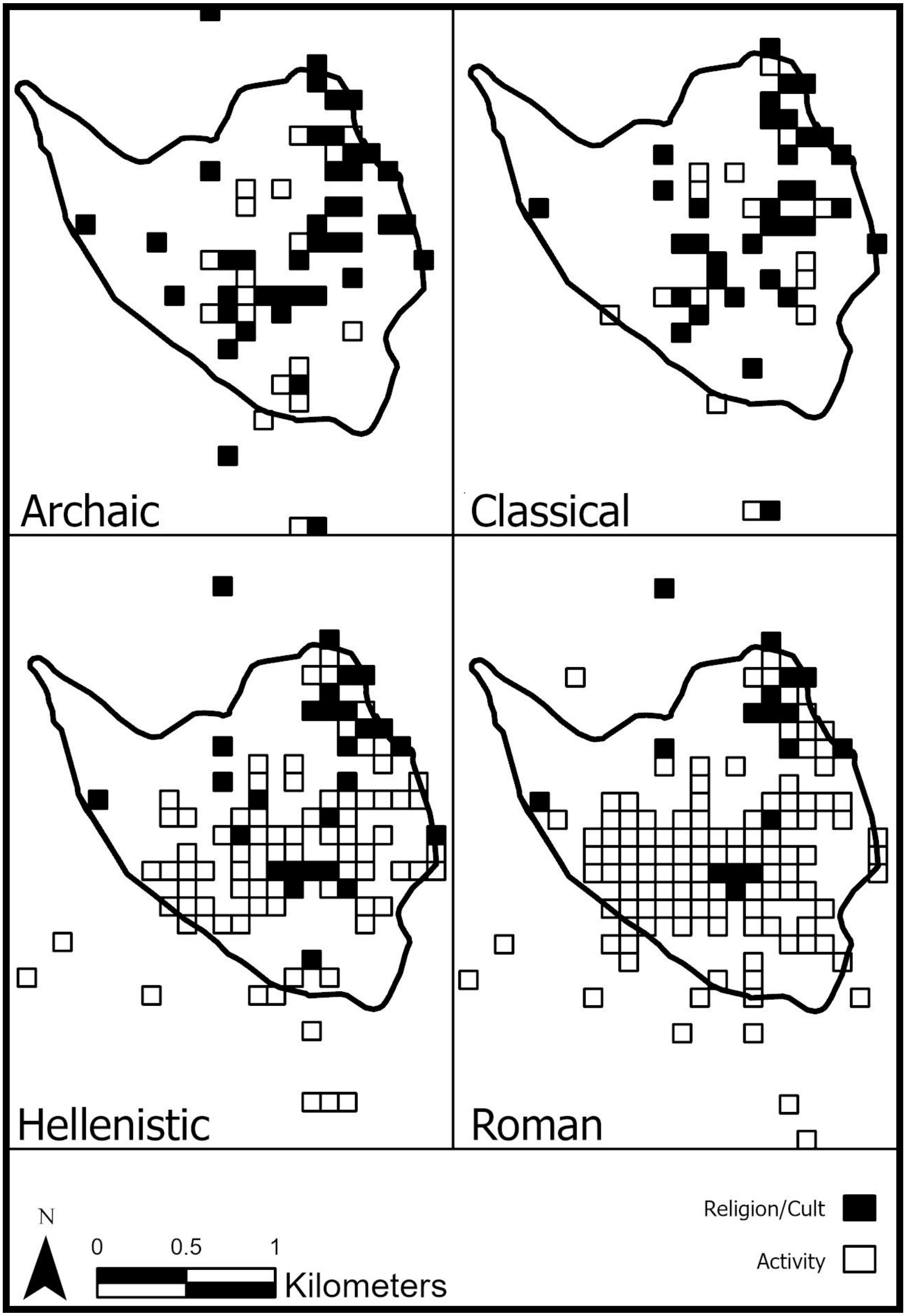

Figure 8: Cult Sites in Sparta, by Period

Table 8: Number of Cult Sites in Athens and Sparta

	# of known cult sites in Athens (other than Acropolis and Agora)	# of known cult sites in Sparta	# of 100-meter tiles in Sparta with cult sites
Archaic period	4	54	43
Classical period	12	42	38
Hellenistic period	3	30	28
Roman period	7	17	17

These caveats are important, but they are not, in our view, grounds to dismiss the obvious conclusions from the data: there were, relatively speaking, a remarkable number of cult sites in Sparta, and the number of those cult sites declined over the course of time. It is tempting to connect the density of cult sites in Sparta to the Spartiates' reputation for piety (see, for example, Flower 2018), and to point to the prominence of sanctuaries in Pausanias' description of Sparta. That approach quite probably has some validity, but it does not explain the long-term decline in cult sites. Further analysis rapidly leads into complicated territory. Suffice it to say for the moment that we believe that the intersection of the socio-political structure of Lacedaemon on one hand and the settlement organization of Sparta on the other was a key factor. As Spartiate lifestyles became progressively more 'normalized' as the result of the diminution of Lacedaemonian territory and autonomy, the need for cult sites declined while the demand for residential space increased, resulting in a significant transformation in the use of space in Sparta.

There are also interesting patterns in the spatial distribution of cult sites. In considering the relevant data, two issues need to be kept in mind. First, the spatial distribution of excavations, both systematic and rescue, in the city has been markedly uneven. As is apparent from Figure 9, many rescue excavations have been carried out in areas immediately to the south and east of the Palaiokastro plateau, whereas other parts of the city remain almost entirely unexplored. Second, the post-depositional histories of different parts of the city vary markedly. The area to the northeast of Palaiokastro was regularly inundated by the Eurotas, which buried votives under protective layers of alluvium and discouraged dense settlement, whereas the area to the south of Palaiokastro was free from flooding and was heavily settled in the Byzantine period. These issues overlap in the sense that they both favor the discovery of votive objects in the northeastern part of the city as opposed to other areas.

Even with those distortions, the data suggest that during the Archaic and Classical periods cult sites were relatively evenly distributed across Sparta's urban fabric. Over the course of the Hellenistic and Roman periods, however, cult sites came to be progressively more concentrated in the northeastern part of the city. If it was indeed the case that the inhabitants of Sparta had progressively less need for cult sites and more need for residential space, the transitioning of spaces from religious to domestic functions would have happened more regularly in those parts of the city that were felt to be more desirable areas in which to live. That process could account for the pattern described above. Supporting evidence for that conjecture can be found in the locations of Roman-era mosaics from domestic spaces: mosaics were strongly associated with luxurious homes, survive well, and figure prominently in excavation reports. As is evident from Figure 10, domestic, Roman-era mosaics are concentrated in the area to the south of the Palaiokastro, which was evidently felt to be more desirable residential space than the northeastern part of the city.

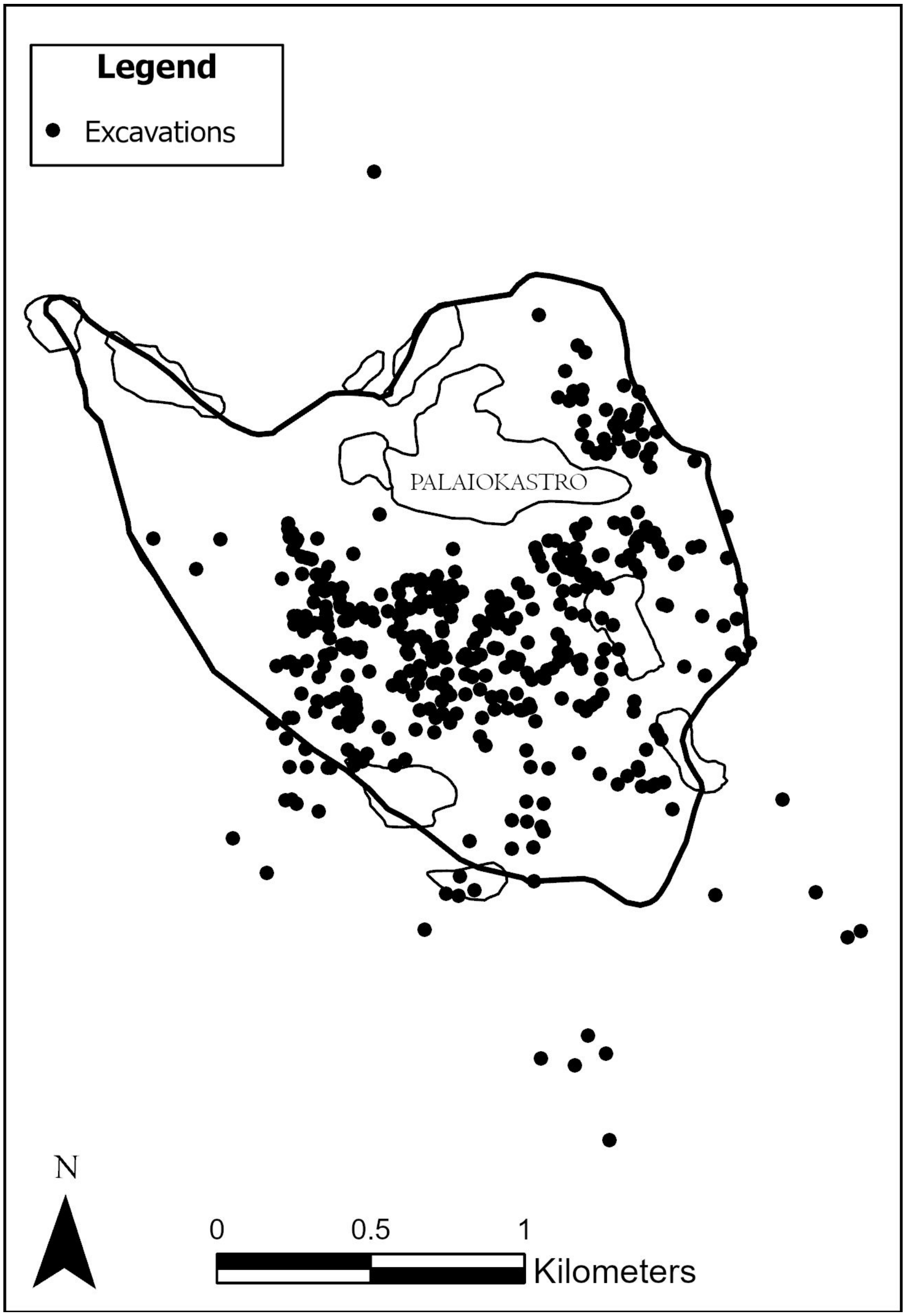

Figure 9: Locations of Reported Rescue Excavations in Sparta

The finds from most of the religious sites in Sparta do not make it possible to identify the specific supernatural being venerated there. It is, however, possible to identify a subset of sites that were almost certainly dedicated to heroes (as opposed to Olympian deities) in the Archaic and Classical periods. As Gina Salapata has shown, terracotta plaques occur primarily at hero shrines in Laconia. For example, the finds from a sanctuary excavated at Stauffert Street include more than 2,500 fragments of terracotta plaques as well as a stone hero relief, whereas only nine such plaques were discovered among the tens of thousands of dedications at the Artemis Orthia sanctuary (c. 200 m south of Stauffert Street) (Dawkins 1929c, 154-5; Flouris 1996; Flouris 2000; Salapata 2014).

Dedications of terracotta plaques in Laconian sanctuaries began in the late sixth century and ended in the late fourth century, so they are a reliable signal only for the Archaic and Classical periods. If we take the presence of significant numbers of terracotta plaques as a reliable signal, 19 of the 54 Archaic and 20 of the 42 Classical cult sites in Sparta were dedicated to heroes.[15] These should be taken as minimum numbers in that they rely on the preservation, excavation, and reporting of a single find type. In terms of spatial distribution, most of the identifiable hero sanctuaries are located to the east and south of Palaiokastro (see Figure 11), with the requisite caveats distortions created by spatial unevenness in excavations in Sparta.

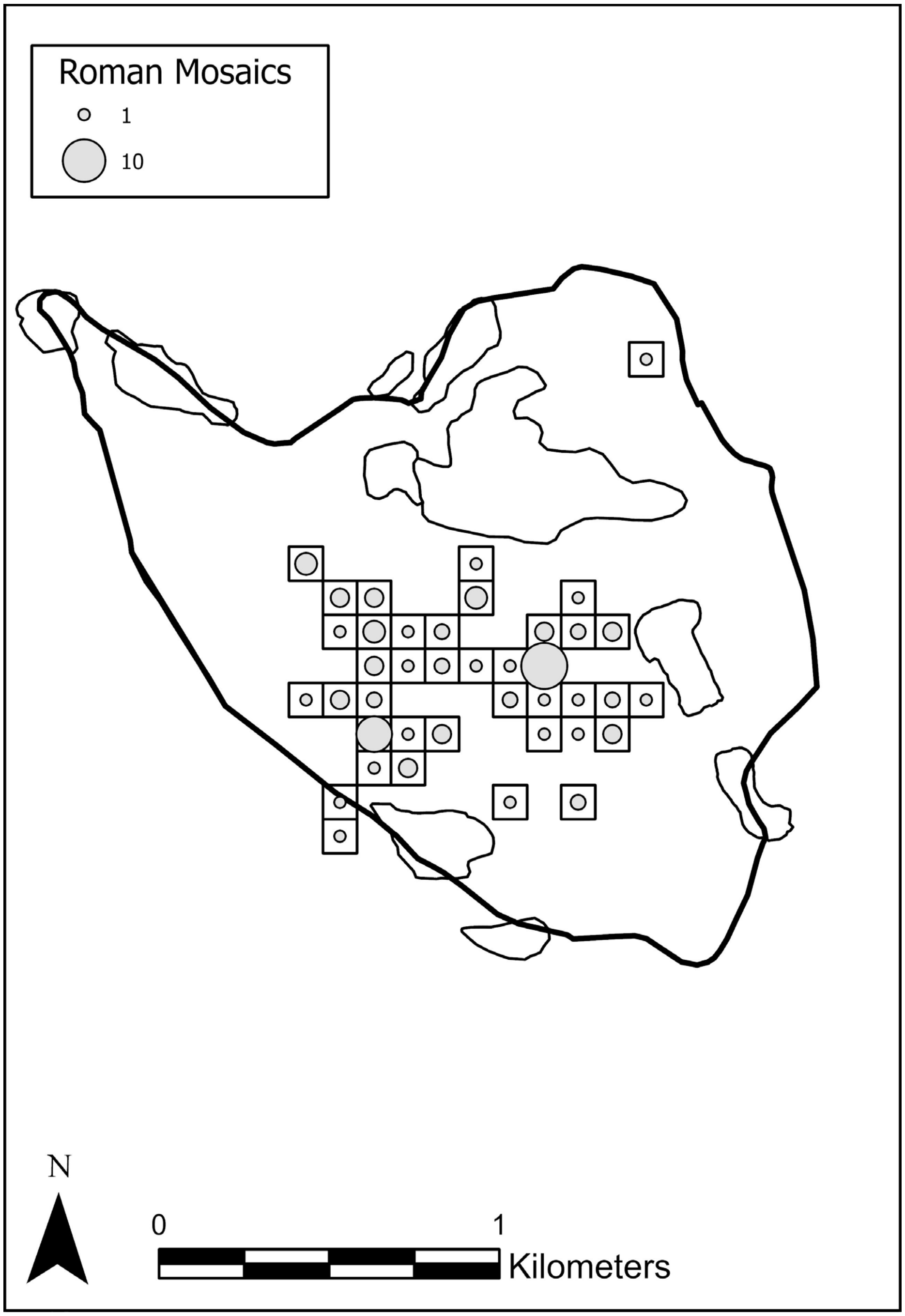

Figure 10: Locations of Roman Mosaics in Sparta

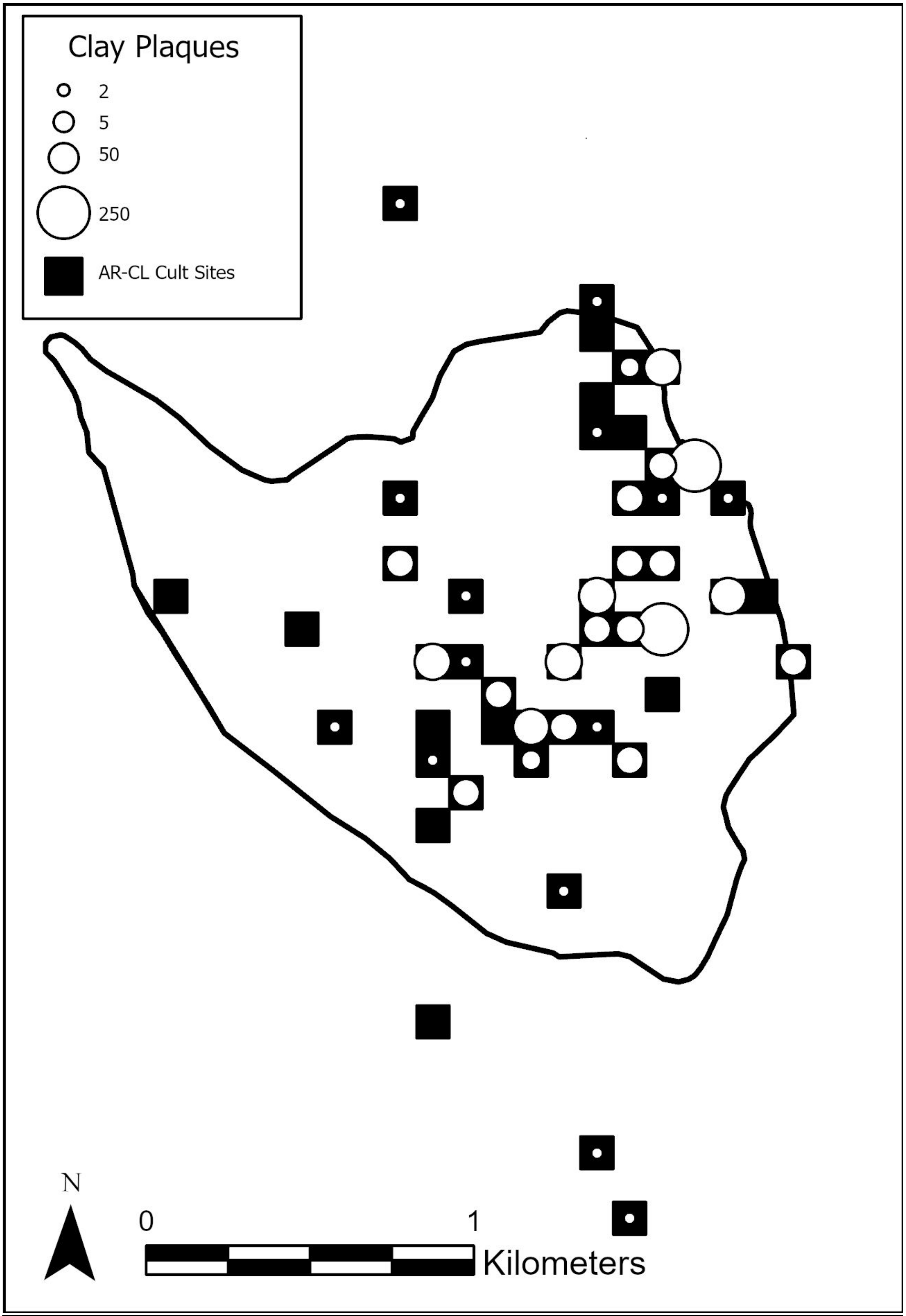

Figure 11: Locations of Archaic-Classical Cult Sites and Finds of Terracotta Plaques

Craft Production and the Extent of Habitation

The recent spate of rescue excavations in Sparta has also produced important evidence bearing on the location of burials and spaces dedicated to craft production and hence to the extent of the inhabited area of the city. There is a widespread pattern, starting in the late Geometric period and continuing thereafter, for burials and craft production to be concentrated on the outskirts of the densely settled areas of Greek communities (see, for instance, Papadopoulos 2003 on the Kerameikos in Athens). That precise pattern does not, of course, hold true in Sparta, where intra-communal burials remained common down through the Roman period. Nonetheless, it is worthwhile considering what we now know about the locations of craft production workshops in Sparta insofar as that would provide insight into the extent of the densely settled part of the city at various points in time. In this case, the evidence is more qualitative than quantitative as it relies on a limited number of sites. We would emphasize at the outset that all of the known production sites in Sparta seem to have been located on the fringes of the densely settled part of the city.

We begin with a site excavated by Dimitris Vlachakos in 2010 on Matala Street in the eastern part of the modern city (BB099; Tile 48.007.051.016; see #1 on Figure 12). Vlachakos uncovered not only the first known remains of a gate in the Hellenistic city wall, but also remains of a pottery workshop in the form of three wells and a circular kiln, active from the late Geometric period to the early Archaic period (Vlachakos 2010). The presence nearby of two Geometric tombs indicates that the area was used for both ceramic production and funerary purposes. Excavations at the Orthia sanctuary, c. 300 m south of this workshop uncovered what the British archaeologists called 'houses' dating to the Classical and Hellenistic periods. The finds from those structures indicate that they were used for craft production, presumably for votives intended for dedication at the sanctuary (Dawkins 1929a, 27-8). Another kiln, this one active in the Classical and Hellenistic periods, was found on the property of K. Riga, c. 500 m south of the Orthia sanctuary (BB108; Tile 48.007.041.050; Soukleris 2014b; Site #2 on Figure 12). All three of the sites are located quite close to the Eurotas river and hence by definition at the eastern edge of the city. They thus fit the aforementioned pattern of production sites being located at the outskirts of Greek urban centers, which in turn suggests that the location of production sites can at least tentatively be taken as an indicator of the spatial limits of the heavily settled areas of ancient Sparta.

In the Archaic period, the Matala Street workshop had a counterpart on the other side of town, at a site located between the Palaiokastro plateau and the Magoulitsa river, c. 200 m east of the line of the Hellenistic city wall (BB140; Tile 48.006.050.026; Zavvou 2004; Site #3). The presence of substantial cemeteries on the eastern edge of the Magoulitsa ravine (Themos et al. 2009; Tsouli 2013a) is a strong indicator that the Magoulitsa was perceived to be an important boundary for the city. The location of this workshop suggests that in the Archaic period the heavily settled area of the city did not yet extend all the way to the Magoulitsa.

In the southwestern part of the city, less than 200 m from the line of the Hellenistic wall and the later Roman Southwest Roman cemetery, excavations uncovered the remains of a Hellenistic pottery workshop (BB034; Tile 48.007.031.062; Zavvou 2006; Site #4). The finds include kiln waste, wedges for supporting pots in a kiln, burnt soil (probably the remains of a kiln), and a well.

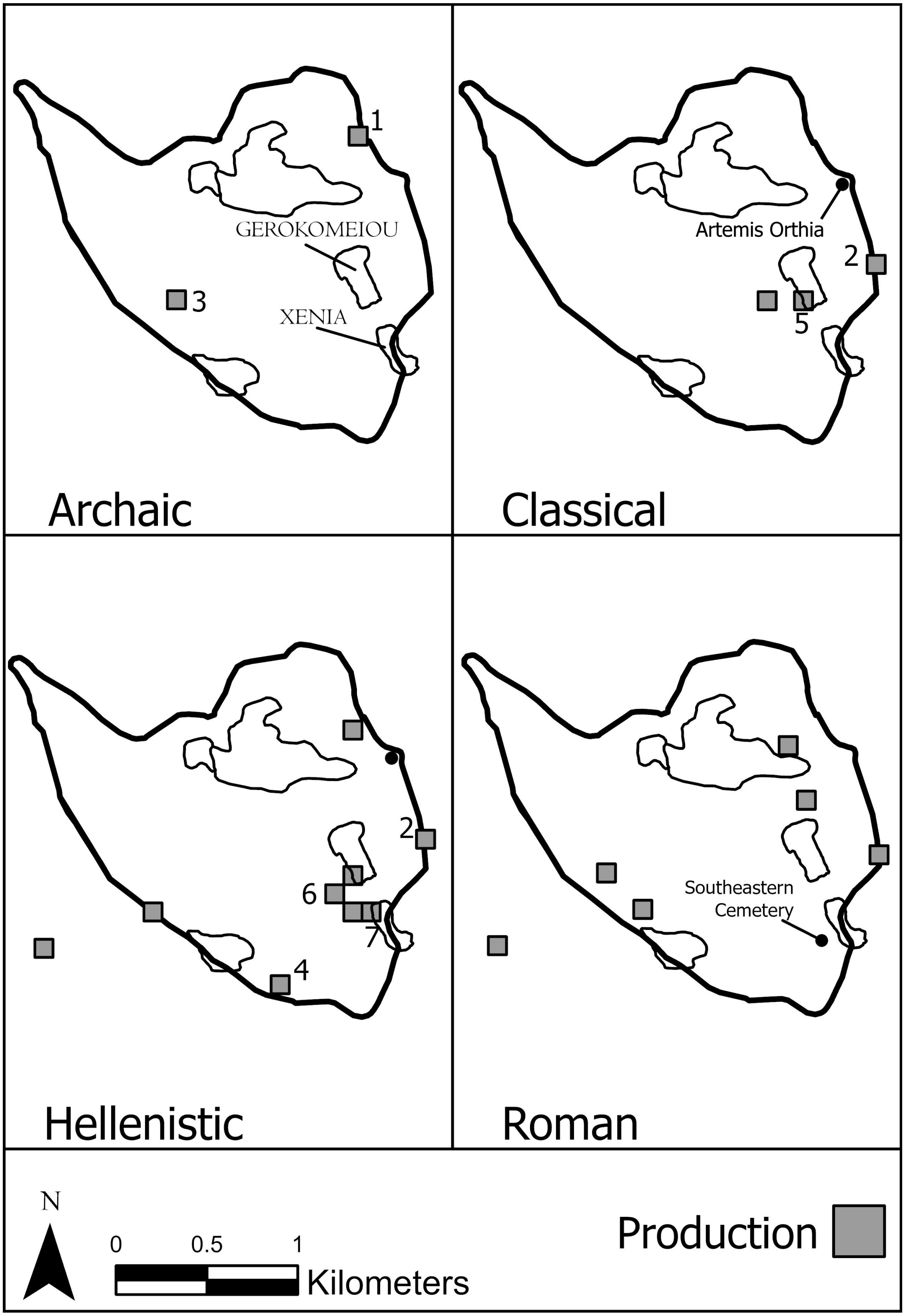

Figure 12: Production Sites in Sparta, by Period

This collection of evidence facilitates interpretation of intriguing finds from the southeastern part of the city. A number of molds were found in Classical-period strata on the southwestern slopes of Gerokomeiou hill, an area which had been used for cultic activity and burials starting in the Archaic period (BB059; Tile 48.007.041.026; Raftopoulou 1998, 134-5; Maltezou 2010a; Site #5). In the early decades of the 20th century, Alan Wace noted the existence, just to the south (on the northern and eastern slopes of Xenia hill), of 'traces of ancient potteries, which consisted of pits whence clay had been dug. They had afterwards been filled up with broken pots and other refuse from the kilns' (Wace 1907, 6). Wace also records that clay was still being dug from this part of town and taken to kilns a little to the southeast, to the area around the Psychiko monument. There must have been substantial clay beds on the slopes of both Gerokomeiou and Xenia hills. Excavations on Gerokomeiou hill also uncovered two pottery kilns dating to the Hellenistic period (BB047; Tile 48.007.041.105; Soukleris 2014a; Site #6). A little to the south, along the northwestern slope of Xenia hill, deformed vases of the Classical and Hellenistic periods, almost certainly discards from a pottery workshop, were found, as well as a mold of the Hellenistic period (BB Π3; 48.007.041.007; Zavvou 1996; Zavvou 2000; Site #7). From that same tile comes what the excavators call a Classical-period '"σαμαρωτός" μυλόλιθος', which is translated into English in the Δελτίον article as 'Egyptian boat quern'. This is presumably a reference to a millstone intended to be operated with a donkey (M.S. 15109). Excavations in the area a little (c. 200 m) to the west of Xenia hill uncovered 46 burials of the Roman period, in what is now sometimes referred to as the Southeastern Cemetery (Tsouli 2020, 153-4).

All of this suggests that Gerokomeiou and Xenia hills were largely reserved for production space starting in the Classical period at the latest and presumably well before that. Moreover, the presence of Roman burials to the west of Xenia hill is likely an indication that this area had at no point been heavily settled, as it seems unlikely that what had been inhabited space was converted to funerary purposes at a time when there was a great deal of building activity in Sparta. The earliest materials uncovered in rescue excavations in this part of the city (BB48, 48A, 49, 49A, 50, 54, 55, 56, 56A) date to the Roman period (Vlachakos and Maltezou 2011; Maltezou 2010b; Maltezou 2011; Tsouli 2013c; Tsouli 2013b; Tsouli et al. 2014). This whole area of town, in the southeastern corner of the Hellenistic city wall, near the clay beds on Gerokomeiou and Xenia hills, could very well have been Sparta's Kerameikos.

That conclusion is reinforced by topographical considerations: it is precisely in this area that the land begins to slope down into the marshy area where the Magoulitsa joins the Eurotas (see Figure 2). This area was, as a result, probably, like the area near the Eurotas, wet and prone to seasonal flooding and hence less than ideal residential space. While one might think that the entire area within the Hellenistic city wall was heavily settled, the course of that wall in the southeastern part of the city was virtually certainly dictated by topography: for military purposes the wall needed to run across the high ground of Xenia and Gerokomeiou hills even though this meant that it ran a few hundred meters to the east of where the heavily settled area of the city ended.

Conclusion

Geographic Information Systems have already shown their usefulness in fieldwork[16], and we hope to have demonstrated here the utility of a spatially-enabled database and GIS in the study and visualization of the settlement organization of Sparta. The preceding discussion is preliminary in two senses: the results presented here are a small subset of what the database can tell us about Sparta, and the existing database could be enhanced.

With respect to the former, we expect to publish in the near future a detailed study of everything that we currently know about the settlement organization of Sparta and how it evolved over the course of time. With respect to the latter, the next obvious step is to bring into the database all of the site plans and photographs published in the entries in the Αρχαιολογικόν Δελτίον pertaining to Sparta. Those plans and photos, ideally high-resolution versions stored in the files of the Greek Archaeological Service rather than the low-resolution versions published in the Δελτίον, would all be spatially located, through a process called georeferencing, so that each plan and photo would appear in the precisely correct location when viewed in the GIS system. That addition to the database would virtually certainly make it possible to produce an accurate plan of the street grid in Roman Sparta, which would in turn make it possible to study and understand known remains with a much higher degree of precision and clarity. Much work remains to be done, but the city of Sparta, long an enigma, is slowly revealing its secrets.

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[1] See Dawkins 1909, 4; Macmillan 1911, xxv for pessimistic comments on the value of further archaeological excavations in Sparta. For an overview of the history of British excavations in Sparta, see Catling 1998.
[2] See Hodkinson 2000, 19-64 for an insightful discussion of the literary sources for Lacedaemonian history, and their limitations.
[3] On Thucydides' views of Lacedaemon, see the articles in Powell and Debnar 2021.
[4] We would like to thank Nicola Nenci for sharing with us a GIS file with 1-meter contours for the terrain in the city of Sparta, which is the basis of the topographical base map in Figure 1 and elsewhere.
[5] Athena Chalkioikos: Spallino 2016, with earlier bibliography. Theater: Woodward 1925; Woodward 1926; Bulle 1937; Waywell and Wilkes 1995; Waywell and Wilkes 1999; Waywell 2002; Roberts and Pickersgill 2003. Archaic (Christou's) Stoa: Christou 1963; Christou 1965; Stavridis 1982-1984; Kourinou 2000, 109-114; Greco 2011, 67-74. Roman Stoa: Traquair 1906; Waywell, Wilkes, Bailey and Sanders 1993; Waywell and Wilkes 1994; Waywell, Wilkes, Bland, Sidell, Wilkinson and Chandler 1997; Del Basso 2022. Round Building: Waldstein 1892; Waldstein and Meader 1893; Waldstein 1894; Waywell and Wilkes 1994, 414-19; Greco, Vasilogambrou and Voza 2009; Έ Εφορεία Προϊστορικών και Κλασικών Αρχαιοτήτων 2015; Voza and Greco 2016; Orestidis, Giannakaki and Vlachou 2020; Orestidis 2020.
[6] Artemis Orthia: Dawkins 1929b; Luongo 2013; Luongo 2015; Luongo 2017a; Luongo 2017b. Heroon: Wace 1906a. Eurotas Altar: Dickins 1906. Stavropoulos sanctuary: Delivorrias 1968a; Delivorrias 1968b; Delivorrias 1969. Achilleion: Dickins 1907; Stibbe 2002.
[7] Leonidaion: Waldstein and Meader 1893. Arapissa baths: Wace 1906b. Altar of Psychiko: Tsouli 2020, with earlier bibliography.
[8] For an introduction to Geographic Information Systems, see Shin, Campbell and Burkhart 2022.
[9] https://mappingancientathens.org/en/home/.
[10] See http://www.ktimatologio.gr/.
[11] This tile system, as noted, covers the entirety of Greece and could be used to report, with a high degree of precision and uniformity, the location of excavations anywhere in the country.
[12] The tombs in the Mousga cemetery are fully published, with clear numbers and dates. As discussed, there are only preliminary publications for the other two cemeteries, which makes it at present impossible to enter the requisite details into the database.
[13] These parameters are consonant with the published details of the two cemeteries, on which see Christesen 2019.
[14] The cluster in the southeastern part of the cemetery is now sometimes referred to as the Southeastern Cemetery (see below). The Hellenistic period shows one major cluster, just to the west of the Roman cluster in that area. The low Global Moran's score for the Hellenistic period is largely the result of the existence of more than 20 burials in the northeastern corner of the city, which produces a considerable degree of dispersion.
[15] Small numbers of terracotta plaques (i.e. 1 or 2) have been found at several sites in Sparta that we have not identified as sanctuaries based on the associated finds. Some of those sites seem to have been workshops, on which see below. In other instances, the most likely explanation is the migration of objects and fills within the city over the course the millennia.
[16] See, for example, https://idig.tips/.